2024 International Conference on Emerging Research in Computational Science (ICERCS)

Sign Comm: A Real-Time Indian Sign Language Recognition System Using Deep Learning for inclusive communication

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2024 International Conference on Emerging Research in Computational Science (ICERCS) | 979-8-3315-3496-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICERCS63125.2024.10895381

***Abstract*— Individuals who are deaf often utilize sign language for communication. In Sign Language, the symbols or motions are arranged grammatically. Every motion is referred to as a sign. Indian Sign Language Recognition using Deep Learning model describes the creation of SignComm, which improves communication for those with hearing loss by translating Indian Sign Language (ISL) motions into text. To identify and interpret ISL gestures in real-time. The paper describes the difficulties the deaf community faces due to the dearth of real-time sign language interpretation services. It suggests a method that uses an Automatic Sign Detection system with deep learning models. The research describes current systems and their shortcomings, such as the lack of real-time translation tools and the difficulties the deaf community faces in obtaining sign language interpretation. Next, it presents the suggested system, SignComm, which has benefits including quick sign language decoding, a user-friendly platform that is easy to use and can foster an inclusive atmosphere in public areas and educational institutions a literature review that examines several deep learning models and their applicability to the project is included in the paper. It emphasizes the potential of the ResNet-50 architecture and the significance of a unique ISL dataset. The benefits of the suggested method, including its accuracy, inclusivity, and capacity to address communication barriers, are highlighted in the paper. Additionally, it outlines the steps involved in the implementation process, such as gathering data, training the ResNet model, and deploying in real time. A user-friendly experience for converting ISL gestures into text is the main goal of the intricate user interface design.**

Keywords*— *Indian Sign Language (ISL), Sign Language Recognition, Deep Learning, SignComm, Automatic sign detection, ResNet, ISL gestures

I. INTRODUCTION

A gesture describes a pattern of body gestures that can be either stationary, moving, or mixed. It is a type of informal communication. People with disabilities, usually those with hearing or speech impairments, use sign language, which comprises visual motions and signals. It is a well-organized code gesture in which each sign has a designated meaning. In addition to letters and numbers, these symbols are utilized for everyday expressions like sentences and greetings. Communication is essential for sharing and expressing information and feelings because it fosters understanding and a sense of intimacy between people. It becomes more difficult for people with disabilities to communicate using natural language. Their favourite communication method is frequently Sign Language [1], [2].

However, non-dives typically struggle to understand sign language due to lacking training or expertise in recognizing the signs. This project aims to bridge the communication gap through real-time Indian Sign Language recognition, raising awareness of ISL in India. ISL employs a mix of one or two hands to represent alphabets, unlike its American counterpart, which uses only one hand. This distinction has led to relatively fewer research and development efforts in this field [6], [13]. The goal of this project is a positive step toward closing the communication gap between ISL users and non-users, particularly in social and professional environments [12],[25]. In the world of communication technology, the Indian Sign Language Recognition using Deep Learning model marks a significant milestone. It presents SignComm, a brand-new mobile application that provides the deaf and hearing-impaired community with an incredibly creative solution by translating ISL motions into text. The introduction of assistive technologies outlines the project's purpose, goals, and the technological revolution it delivers to the region

The Indian Sign Language Recognition model is a significant technological milestone. The application, SignComm, provides an innovative solution for the deaf and hearing-impaired community by translating ISL gestures into text. The project outlines the technological advancements and aims to help those with hearing disabilities who face challenges due to the lack of real-time sign language interpretation systems [5],[28]. By leveraging deep learning models, the system process ensures smooth communication with non-users [22],[25].

SignComm integrates a deep learning model, with the ResNet-50 architecture core of the system. This architecture is well-established for tasks such as image recognition and allows the application to recognize and interpret ISL gestures accurately. The application's training process involves a custom dataset of ISL gestures to ensure a robust and reliable translation system [24], [27]. The development of SignComm is a collaborative effort, building on extensive research technologies like Microsoft Kinect, Leap Motion, and Open

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Pose, which have contributed to gesture recognition software advancements [4],[9],and [19]. Unlike other solutions, SignComm distinguishes itself by integrating these technologies into a mobile application accessible to a wider audience [12], [22].

The user-centric design of SignComm reflects the project's commitment to inclusivity. It caters to users with varying leniency in ISL, ensuring that the technology is both advanced and approachable. The intuitive interface makes it easy for users to navigate, ensuring that ISL users can communicate effectively with non-users [6], [28].

II. LITERATURE REVIEW

Integrating Media Pipe with Long Short-Term Memory (LSTM) networks is a major step forward in the recognition of Indian Sign Language (ISL). This method integrates the sequential data processing capacity of LSTM with the robust feature extraction capabilities of Media Pipe, enabling precise comprehension of ISL gestures. A crucial study conducted by Manish Jha, Hardik Pithadiya Rajawat, Hartik Suhagiya, and Professor. Lakshmi Kurup has made a significant contribution to this field through the development of a system capable of recognizing isolated words in ISL. The system underwent evaluation using both the INCLUDE-50 and the broader INCLUDE dataset, attaining accuracies of 94. 8% and 87. 4% respectively. These results exceed the present state-of-the-art performance on the aforementioned datasets. The study also introduced a real-time ISL recognition system for 24 dynamic signs utilizing Media Pipe and LSTM, attaining a testing accuracy of 97%. This underscores the system's capacity for real-time recognition, which is essential for enabling smooth communication within the deaf and mute community. The objective of these systems is not solely confined to gesture recognition but extends to encompass translation into text and speech to facilitate communication between individuals. Potential improvements in the future could involve the expansion of the gesture vocabulary, enhancement of robustness in diverse settings, and implementation of multilingual support to broaden the reach of a more extensive audience [1].

The project designed by Sakshi Lahoti, Shaily Kayal, Sakshi Kumbhare, Ishani Suradkar, and Vikul Pawar is intended to tackle communication obstacles encountered by individuals from the deaf and mute community. This will be achieved through the development of a portable mobile solution catered towards users of American Sign Language (ASL). This application employs smartphones to capture images using the smartphone camera, subsequently utilizing skin segmentation in the YCbCr colour space to separate hand gestures. Feature extraction is conducted utilizing Histogram of Oriented Gradients (HOG), while the SVM classifier is employed for sign interpretation. The system guarantees mobility and accessibility, thus allowing users to engage at any moment and from any location. This methodology illustrates the capability of machine learning in improving pattern recognition and real-time American Sign Language (ASL) translation [2].

In the research entitled Convolutional Neural Network Hand Gesture Recognition for American Sign Language, the authors investigated the utilization of Convolutional Neural Networks (CNN) in identifying hand gestures in American Sign Language. This study focuses on addressing the communication obstacles encountered by the deaf community by utilizing advancements in computer vision and deep learning. The system, which has been devised to identify digits ranging from 0 to 9, employs Convolutional Neural Networks (CNNs) for the processing of acquired images. The system's effectiveness is improved by converting RGB data to grayscale, thereby decreasing storage needs and training duration. With a dropout layer included, an optimized 10-layer architecture achieved 91. 37% training accuracy and 87. 5% testing accuracy. By demonstrating how Convolutional Neural Networks (CNNs) can be used to create inclusive communication tools through portable and real-time Sign Language detection, this study advances the field [3].

A real-time system for translating American Sign Language (ASL) into text has been developed by Olayah, Mohammed Al Yami, and Yahya A. Abdelrahman. This system recognizes ASL motions by combining advanced image processing and machine learning techniques. The CNN model analyses spatial hierarchies present in images, which are crucial for proficient gesture recognition. The researchers have ensured that the system operates effectively across diverse environmental conditions, encompassing lighting, background, and individual signing styles. This method emphasizes the capability of machine learning techniques in developing inclusive communication tools that eliminate obstacles for the deaf community [4]. The comprehensive review conducted by Noraini Mohamed, Mumtaz Begum Mustafa, and Nazean Jomhari examines the progress in vision-based hand gesture recognition between 2014 and 2020. The paper examines the advancements and deficiencies in the field, the comprehensive review conducted by Noraini Mohamed, Mumtaz Begum Mustafa, and Nazean Jomhari examines vision-based hand gesture recognition between 2014 and 2020. The paper examines the advancements and deficiencies in the field, with a specific focus on elements such as data collection, environmental variables, and the depiction of hand gestures. The review categorizes recognition accuracy into two groups: signer-dependent, with a range of 69% to 98%, and an average percentage. The accuracy rates range from 48% to 97%, with an average of 88. 8%, and are not dependent on the signer. Seventy-eight-point two percent (78. 2%). The authors underscore the necessity of ongoing gesture recognition and the development of more resilient approaches for practical use, particularly in the fields of assistive technology and human-computer interface (HCI) [5].

The paper titled MobileNetV2: Inverted Residuals and Linear Bottlenecks authored by Mark Sandler, and other co-authors presents the MobileNetV2, a sophisticated neural network structure designed specifically for enhancing performance in mobile and embedded vision tasks. MobileNetV2 enhances the original Mobile Net model by implementing inverted residuals and linear bottleneck layers, thus increasing efficiency while maintaining accuracy. This architectural design achieves a harmonious equilibrium among latency, memory consumption, and precision, rendering it well-suited for real-time image recognition functionalities on mobile devices. The experimental findings indicate that MobileNetV2 surpasses alternative lightweight models in conventional benchmarks, rendering it a viable option for mobile visual tasks [6]. In the study titled Towards Continuous Indian Sign Language Recognition Using Convolutional Neural Networks and Recurrent Neural

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Networks, V. S. Chauhan, R. Kumar, and V. K. suggest a hybrid deep learning method that successfully captures the spatial and temporal features of ISL gestures by combining Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). The problem of recognizing continuous gestures, which are crucial for organic communication, is addressed by this paradigm. The results show significant improvements in recognition accuracy that help with real-time ISL interpretation. This model offers a more robust approach to continuous sign language recognition by leveraging the advantages of both Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) [7].

The authors V. S. Chauhan, R. Kumar, and V. K. Singh also investigate the utilization of a modified VGG16 model for the real-time recognition of Indian Sign Language gestures. By utilizing the VGG16 architecture, the authors have enhanced the system's capacity to accurately and swiftly recognize ISL gestures. The adapted VGG16 effectively captures spatial features, thus improving communication accessibility for the deaf community through real-time sign language translation. This adaptation allows the model to function effectively even in environments with limited resources [8].

In the publication titled Real-Time Hand Gesture Recognition based on deep learning YOLOv3 Model, a hand gesture recognition system employing the YOLOv3 and DarkNet-53 Deep Learning architectures is presented. The system attains a 97. 68% accuracy in the Identification of static gestures. The researchers propose that the system be improved for multi-gesture detection and Utilized in assistive living systems catering to both healthy and disabled individuals. The YOLOv3 model demonstrates superior performance over alternative models in hand feature extraction, precision, recall, and F-1 score, positioning it as a promising solution for real-time gesture recognition as cited by reference [9].

In the article titled A Survey of Recent Advances in Hand Gesture Recognition, the authors conduct an analysis of prevailing trends and obstacles encountered in gesture recognition systems. Special emphasis is placed on machine learning models and their utilization in real-time systems. The survey encompasses a broad spectrum of recognition Systems, ranging from conventional computer vision methodologies to deep learning strategies. It assesses the efficacy of diverse systems in diverse environments. The text also emphasizes critical areas for future research, such as enhancing gesture recognition in diverse lighting conditions, increasing resilience to noise, and minimizing system latency to achieve quicker responses. This thorough review serves as a roadmap for future research in the field and proposes potential avenues for innovation, such as hybrid models amalgamating classical and deep learning techniques to enhance hand gesture recognition [10].

In the research article titled Towards Continuous Indian Sign Language Recognition Using Convolutional Neural Networks and Recurrent Neural Networks, authored by V. S., Chauhan, R. Kumar, and V. K. Singh has put forth a hybrid deep learning strategy that combines recurrent neural networks (RNNs) and convolutional neural networks (CNNs) to capture the temporal and spatial characteristics of ISL gestures. The difficulty of correctly recognizing continuous gestures which are crucial for impromptu communication is addressed by this paradigm. The results demonstrate significant improvements in recognition accuracy, which improves real-time ISL interpretation. This model offers a more robust method for continuous sign language recognition by fusing the advantages of recurrent neural networks (RNNs) and convolutional neural networks (CNNs) [11].

Mr. P. P. Roy, Esq. Chaudhury and S. Das proposed a technique for the recognition of Sign Language through the usage of transfer learning with deep convolutional neural networks (CNNs), bolstered by the inclusion of spatiotemporal characteristics. This approach capitalizes on utilizing pre-trained models trained on extensive datasets, adapting them to the particular objective of sign language recognition, thereby mitigating issues related to limited data availability and model adaptability. Through the incorporation of spatiotemporal features, the system adeptly captures the spatial arrangement of the hand and the temporal characteristics of gesture motion, thereby enhancing accuracy and robustness in the recognition of dynamic sign gestures [12].

V. S. Chauhan, R. Kumar, and V. K. Singh further investigate the utilization of a modified VGG16 model for the real-time recognition of Indian Sign Language. By adopting the VGG16 architecture, the authors enhance the system's capability to accurately and swiftly recognize ISL gestures. The adapted VGG16 efficiently captures spatial features, thereby improving communication accessibility for the deaf community via real-time sign language translation. This modification allows the model to operate effectively even in environments with limited resources [13].

The use of 3D Convolutional Neural Networks (CNNs) for Indian Sign Language (ISL) detection was studied by N. Gupta, S. Verma, and A. Chaudhary. Unlike traditional 2D CNNs, 3D CNNs are able to capture both spatial and temporal features in video data, which makes them ideal for recognizing dynamic hand motions in continuous sign language. Their model performed well on trials involving both static and dynamic sign recognition, and it demonstrated impressive accuracy in identifying ISL movements. The study emphasizes how important it is to use temporal features to improve the fluidity and authenticity of sign language recognition [14].

The authors Muneer Al-Hammadi, Ghulam Muhammad, Wadood Abdul, and Mansour Alsulaiman have proposed a deep learning-based methodology for the recognition of Sign Language gestures, emphasizing the efficient representation of hand gestures. The proposed system enhances hand gesture features through the utilization of deep learning models to enhance recognition accuracy and efficiency. The technique employs feature extraction methods to improve the identification of hand postures, thereby guaranteeing resilience against fluctuations in gesture velocity and background interference, rendering it appropriate for real-time applications [15].

S. Subburaj and S. Murugavalli present an in-depth analysis of vision-based and Deep-learning methodologies for the recognition of sign language. The survey encompasses a broad spectrum of methodologies, ranging from conventional computer vision methods to sophisticated deep learning models, scrutinizing their efficacy within the realm of sign language recognition. It examines obstacles such as the collection of vast amounts of data, real-time processing, and the incorporation of temporal dynamics into gesture

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recognition. This review serves as a valuable asset for researchers who are endeavouring to enhance the efficiency and precision of sign language recognition systems [17].

Ao Tang, Ke Lu, Yufei Wang, Jie Huang, and Hougang Li have engineered a deep neural network-based real-time hand posture recognition system. The system employs a blend of convolutional layers to capture intricate features of hand postures, thereby attaining a high level of recognition accuracy in real-time applications. The authors emphasized the significance of utilizing deep learning techniques for posture recognition due to their ability to enable system adaptation to diverse users and environmental conditions, rendering the system applicable for real-world scenarios [18].

Sakshi Sharma and Sukhwinder Singh examined the application of vision-based deep learning methodologies for the identification of hand gestures. Their work showcases the capabilities of convolutional neural networks (CNN) in interpreting sign language gestures across diverse environments. By training the model using extensive gesture datasets, significant enhancements in recognition accuracy were attained, thereby rendering the system more efficient for real-time applications involving the interpretation of sign language [19].

Neethu, R. Suguna and Divya Sathish have presented an effective technique for detecting and recognizing human hand gestures utilizing convolutional neural networks (CNNs). Their methodology integrates hand detection and gesture recognition within a unified framework, enhancing the utilization of deep learning for applications requiring real-time processing. The model exhibited strong performance in dynamic gesture recognition scenarios, positioning it as a viable solution for applications in human- computer interaction and assistive technologies tailored to the needs of the deaf community [20].

Abdullah Mujahid, Mazhar Javed Awan, Awais Yasin, Mazin Abed Mohammed, Robertas Damaševičius, and Rytis Maskeliūnas put forward a real-time hand gesture recognition that is founded on the YOLOv3 model. The system attains precise gesture recognition through the utilization of YOLOv3's object detection capabilities, allowing it to identify and acknowledge hand gestures in real-time. This approach presents notable enhancements in both precision and computational velocity, rendering it appropriate for practical scenarios where instantaneous identification is crucial [21].

The authors Pulkit Rathi, Raj Kumar Gupta, Soumya Agarwal, and Anupam Shukla utilized the ResNet50 architecture for sign language recognition. Through the implementation of a deep residual network, their system achieved a higher level of accuracy in recognizing both static and dynamic signs. The utilization of residual blocks serves to mitigate vanishing gradient issues, thereby enhancing the efficiency of training deeper models. The system exhibited a high level of accuracy on conventional sign language datasets, thereby making a significant contribution to the progress of deep learning-enabled sign language recognition [22].

The individuals named Tanseem N. Abu-Jamie and Samy S. Abu-Naser's study entitled Classification of Sign-Language Using Deep Learning by ResNet (2022) investigates the efficacy of ResNet, a robust deep learning framework, in the classification of sign language gestures. The study underscores the potential of Residual Networks (ResNet) to address the vanishing gradient issue in deeper networks, thereby enhancing performance in Sign Language recognition tasks. Through the utilization of this methodology, the model attained enhanced precision in the identification of both static and dynamic gestures, rendering it appropriate for real-time sign language recognition systems. The study underscores the benefits of ResNet in managing intricate sign language datasets, presenting a promising pathway for forthcoming Sign Language recognition systems [24].

Pratik Likhar, Neel Kamal Bhagat, and Rathna G. N. (2020) conducted a study on different deep learning methodologies aimed at recognizing Indian Sign Language (ISL). Their research delves into the potential of Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTMs), and hybrid models for effectively classifying International Sign Language (ISL) gestures. Their focus was directed towards the development of an effective system capable of operating in real-time, regardless of fluctuating lighting and environmental circumstances. Their system attained a high level of recognition accuracy, and they additionally investigated strategies to enhance robustness by integrating hand shape and motion features into the model. The authors have concluded that the integration of deep learning models with resilient feature extraction techniques results in notable enhancements in the efficiency of ISL recognition systems [25].

In the research conducted by Sruthi C. J and Lijiya A. In their study titled A Deep Learning-based Indian Sign Language Recognition System published in 2019, the authors concentrated on utilizing deep neural networks for the recognition of Indian Sign Language The system utilizes a Convolutional Neural Network architecture to analyze and categorize static hand gestures in the context of International Sign Language (ISL), attaining a notable level of precision in the recognition of diverse gestures. By integrating preprocessing methods such as image resizing, noise elimination, and normalization, the system adeptly addresses obstacles such as background confusion and variations in lighting. Additionally, the system has demonstrated its potential for use in real-world scenarios, including interactive devices and mobile applications that let people with hearing impairments communicate with the broader public [26].

In 2019, Neel Kamal Bhagat, Y. Vishnusai, and G. N. Rathna proposed a deep learning and image processing approach for recognizing gestures in Indian Sign Language. Their system leverages a blend of conventional image processing methods for hand detection and advanced deep Learning models such as CNNs for recognizing gestures. The study emphasizes the significance of integrating traditional methodologies with contemporary deep learning techniques to enhance recognition precision. They also illustrate the efficacy of their model in diverse environmental settings, highlighting its suitability for real- time communication within the deaf and hard-of-hearing community [27].

The research conducted by Harsh Kumar, Vashisth, Tuhin Tarafder, Rehan Aziz, Mamta Arora, and Alpana (2023) is centred on the use of Deep Learning methodologies for hand gesture recognition in Indian Sign Language (ISL). Their research delves into the utilization of Convolutional Neural Networks (CNNs) and transfer learning techniques using pre-trained models like VGG16 and ResNet for the recognition of

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ISL hand gestures. The model showed notable enhancements in recognition accuracy through the integration of diverse preprocessing techniques, such as manual segmentation and extraction of features from multiple perspectives. This study contributes to the improvement of real-time hand gesture recognition systems for ISL, intending to facilitate a smoother interaction between users and devices (28).

R. Dhiman,G. Joshi and Rama Krishna (2021) presented an advanced deep-learning methodology for the classification of ISL gestures, which considers various background conditions. Their work is centered on the training of deep learning models using a wide range of images featuring diverse backgrounds in order to enhance the robustness of the model. The researchers employed a blend of Convolutional Neural Networks (CNNs) and data augmentation methodologies to address background interference and guarantee precise identification in diverse settings. This approach is essential for the practical implementation of ISL recognition systems, as the background may vary considerably across different settings.

Soumen Das, Saroj Kr Biswas, and Biswajit Purkayastha (2022) put forward a method for automatizing the recognition of Indian Sign Language. This was achieved by integrating deep learning features with manually crafted features. Their methodology combines the advantages of deep learning, known for its proficiency in acquiring high- level characteristics, with handcrafted features, designed to capture intricate low-level details that are essential for precise recognition. By amalgamating these two types of attributes, the system demonstrated enhancements in classification precision. The authors have shown that this hybrid method is particularly advantageous for categorizing signs exhibiting subtle variations, a phenomenon frequently encountered in ISL. The creation of more accurate and Adaptable ISL recognition systems is made possible by this study.

III. PROPOSED SYSTEM

The proposed system, SignComm, is a mobile application designed to translate Indian Sign Language (ISL) into text in real time It aims to address communication barriers faced by the deaf community by providing instant translation of ISL gestures The system utilizes an Automatic Sign Detection mechanism powered by deep learning models, specifically ResNet-50, to recognize and interpret ISL gestures The application's user-friendly interface allows for easy interaction and displays the translation promptly, facilitating clear communication.

SignComm's real-time translation capability bridges the gap between the deaf community and society, enhancing inclusivity in public spaces and educational settings. The system's architecture integrates advanced technologies such as machine learning, computer vision, and mobile app development to create an accessible and efficient platform for ISL users to express themselves confidently6. The project represents a significant step towards an inclusive society where communication barriers are minimized, and diverse linguistic abilities are embraced.

System Architecture Enhancements in the context of the proposed system, a pivotal focus lies on the enhancement of system architecture to embrace modern, scalable, and modular principles. This architectural refinement is instrumental in ensuring adaptability to future technological trends, facilitating the seamless integration of new features, and ultimately contributing to improved overall system performance. The architecture for the SignComm Indian Sign Language Recognition model is based on a deep learning approach, specifically employing Convolutional Neural Networks (CNNs). The model is designed to process images of hand gestures in real-time, which are the primary means of communication in sign language principles. The architecture for the SignComm Indian Sign Language Recognition model is based on a deep learning approach, specifically employing Convolutional Neural Networks (CNNs). The model is designed to process images of hand gestures in real-time shown in Fig.1, which are the primary means of communication in sign language.

A. ResNet 50 Integration

In this model ResNet-50 is used for image recognition tasks it incorporates shortcut connections (skip connections) to address the vanishing gradient problem.

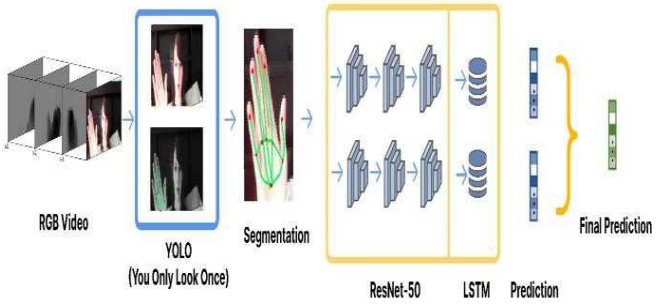


Fig. 1. Architecture of Proposed Sign comm

Fig. 1. Architecture of Proposed Sign comm

The architecture includes bottleneck blocks and achieves high accuracy without excessive computational cost by adapting ResNet-50 for video-to-text hand sign classification, the architectural design encompasses components for data preprocessing, model training, and real-time inference.

1) Data Preprocessing Module

The data preprocessing module handles video frame extraction hand region detection, image preprocessing, feature extraction, Dataset splitting, and data loading pipeline. Considering the nature of the proposed system, the data preprocessing module comprises functions such as sign identification and translation.

2) ResNet50 Model Training

Model A training dataset of images regarding hand signs along with labels was used to train the ResNet50 model. The pre-processed images are of the same size and format. The dataset has been divided into training, validation, and testing sets. The last fully connected layer is modified to output the number of classes in the dataset. Then the model is further fine-tuned by freezing earlier layers and training only the added layers with the dataset. After the training, the model's performance is evaluated in Real-time Inference Component.

3) Real-time Inference Component

The real-time inference component of the architecture enables real-time translation for hand sign gestures. Users would record videos while the system employed fine-tuned ResNet50 for simultaneous translation. The modular design ensures scalability so that the system can handle high variability in workload, yet the system need not compromise on real-time responsiveness

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4) Scalability and Integration

The Architectural improvements can focus on scalability as the ResNet50 module is integrated into the whole system with no constraints. The model is implemented as a RESTful API, where it would allow integration with the mobile applications to be offered to users in order to allow them to interact with the model through HTTP requests. Scalability can be achieved through by using parallel processing techniques.

B. Integration of Emerging Technologies

1) Technological Framework

The technological framework behind real time Indian Sign Language (ISL) detection using Deep Learning in Res Net, integrated within a Flutter mobile application, is designed to efficiently recognize and translate ISL gestures. At its core, this framework utilizes state-of-the-art Deep Learning models, specifically Res Net, renowned for its image recognition capabilities. These models are meticulously trained on a diverse dataset of ISL gestures, enabling accurate identification of complex hand movements and facial expressions inherent in ISL Integrating Res Net-based ISL recognition into a Flutter mobile app provides user-friendly access, allowing live video feeds or images to be swiftly processed through the Deep Learning model in real-time. The backend infrastructure supports the computational demands of the Deep Learning model, handling the translation process by mapping recognized ISL gestures to corresponding English equivalents using a comprehensive database and advanced algorithms This amalgamation not only promotes inclusivity but also bridges communication gaps between linguistic communities.

2) User Interface Design

The application's user interface (UI) design is crafted to facilitate seamless interaction with real-time sign language translation. The primary screen features of application shown in Fig.2 a prominent Record button, initiating the camera view for capturing sign language gestures Upon activation, the UI transitions to the camera view, translating gestures into English text displayed in a designated field a Stop button ends recording, compiling the text into a coherent sentence displayed in the text field Users can copy this sentence to the clipboard for sharing or storage This UI prioritizes simplicity and ease of use, enabling effective cross-modal interactions between Indian Sign Language and English.

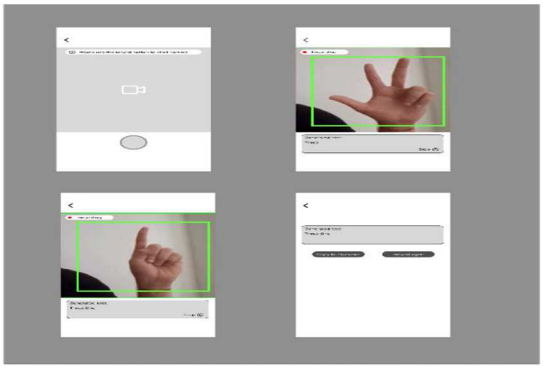


Fig. 2. Sign Comm Application Interface

3) Integration of Automatic Sign Detection

Detection integrating automatic sign detection enriches the user experience by leveraging computer vision algorithms and machine learning models to recognize and interpret ISL gestures in real- time. The application uses convolutional neural networks (CNNs) trained on ISL datasets to identify and classify hand movements and expressions as users present ISL gestures to the camera, the application translates them into English text almost concurrently, fostering inclusive communication between different linguistic communities

4) Testing and Validation

Testing ensures the applications reliability, accuracy, and usability Functional testing validates core functionalities like navigation and sign detection accuracy Performance evaluation includes stress testing to assess real- time video feed processing. Usability testing gathers feedback from ISL speakers to refine user experience Validation involves comparing translations against known ISL gestures for accuracy Continuous testing and user feedback drive iterative improvements to enhance the applications performance and user experience.

5) Workflow

The workflow of our real-time Indian Sign Language (ISL) translation and gesture recognition application begins with an intuitive user interface design tailored for seamless interaction. On the main screen, most prominently displayed is a Record button - via which users can activate the camera view for capturing ISL gestures shown in Fig.3. When activated, users find live video feeds on the screen to present their sign language expressions directly to the device; our application dynamically translates these gestures into English text, rendering them as individual text segments appearing in a particular field on the screen for the recording session may be concluded at one's convenience by clicking the convenient Stop button, which triggers the application to join the sequence of translated text segments into an intelligible sentence. At the end of a recording session, the users simply click on the strategically located Stop button, which provokes and after that, the application collects the translated word pieces to form a meaningful sentence. Collecting the compiled sentence into its integral form of the intended message, then into the text box. More importantly, users could paste this compounded sentence directly to the clipboard for easy sharing and storage because they may need it in the future. This compiled sentence, representing the totality of the message communicated, is then displayed inside the prescribed text Field.

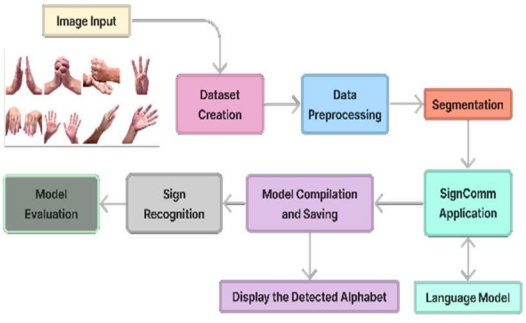


Fig. 3. Workflow of Sign comm

Automatic sign detection through smooth integration into our application is another very important feature. It increases functionality and quality in terms of user experience computer vision.

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The application is set with algorithms and machine learning models that enable the automatic identification and interpretation of the ISL gestures presented to the camera in real time. This is comprised of using CNNs or other deep learning architectures, for which tremendous attention shall be put on huge sets of ISL gestures with massive subletting in the hand movement and expression to strongly detect and classify them. Testing and validation processes play a vital role in ensuring the reliability, accuracy, and usability of our application designed for real-time ISL translation using automatic detection has shown in Fig.4.



Fig. 4. User Interface of the SignComm

Functional testing validates core functionalities such as navigation and sign detection accuracy, ensuring seamless operation. Performance evaluation includes stress testing to assess real- time video feed processing without compromising performance. Usability testing gathers feedback from ISL speakers to refine the user interface and translation accuracy, while validation compares translations against known ISL gestures for precision and correctness Continuous testing, validation, and iterative enhancements based on user feedback ensure that our application evolves to meet the evolving needs and expectations of its users.

IV. Conclusion

The SignComm technology, which allows for the real-time translation of Indian Sign Language (ISL) into text, is a major step forward in closing communication gaps between the deaf population and society. Gesture detection is made more accurate and efficient by combining computer vision techniques with deep learning models, such as ResNet-50. The user-friendly mobile application guarantees smooth interaction and promotes inclusion because it is based on a strong technological foundation. The model's capacity to handle a variety of dialects, the app's performance optimization for quicker real-time processing, and the expansion of the ISL dataset for improved gesture detection could be the main areas of future improvements. The user experience might be further enhanced by adding voice feedback and multilingual support, which would increase SignComm's adaptability and accessibility for users worldwide.

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